Varun Satish

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Education

University of Illinois Urbana-Champaign August 2022 - May 2026
Bachelor of Science, Astrophysics + Data Science with Distinction

- Honors: Distinguished Undergraduate Researcher Certificate; Astronomy Graduation Speaker
- Thesis: *Exploring Escape Fractions in ENZO Simulations of the Epoch of Reionization*

Research Experience

Stellar Modelling/Cosmology Research July 2026 – Present
Post Baccalaureate Researcher

- Leading computational research projects investigating the effects of stellar initial mass functions (IMFs) on galaxy evolution in ENZO cosmological simulations.
- Using Flexible Stellar Population Synthesis (FSPS) for stellar population modeling and validate collisional reaction rates for a new chemistry solver and radiation network.

Computational Astrophysics and Radiative Transfer Research August 2023 – May 2026
Undergraduate Researcher

- Led an independent research project investigating ionizing radiation escape fractions in ENZO simulations of the Epoch of Reionization that was submitted to The Astrophysical Journal.
- Co-developed a numerical radiative transfer pipeline incorporating ray tracing to calculate optical depths around individual star particles, accounting for local gas properties and ionization parameters.

Astronomy Education Research January 2025 – May 2026
Undergraduate Researcher

- Investigated astronomy curriculum and instructional practices in rural Illinois high schools through qualitative analysis of semi-structured teacher interviews.
- Contributed to a nationwide study of astronomy course availability by coordinating outreach to education departments across all 50 states and analyzing changes in public high school course offerings

Publications

Published

- Nguyễn, T. H., Barrow, K. S. S., Byrom, L., **Satish, V.** (2025). "Effect of molecular hydrogen self-shielding modeling on early Reionization Era galaxies in radiative hydrodynamic cosmological simulations." *MNRAS*. https://doi.org/10.1093/mnras/staf2045

Submitted

- Wilken, S., Looney, L. W., Talafian, H., **Satish, V**. "What gets Taught? An exploration of astronomy content and curriculum development in rural high schools (Submitted: APS Physical Review PER).
- Barrow, K. S. S., et al. (including **Satish, V.**) "The AGORA High-resolution Galaxy Simulations Comparison Project. XI..." (Submitted: The Astrophysical Journal). https://doi.org/10.48550/arXiv.2605.24097
- **Satish, V.,** Barrow, K. S. S., Byrom, L., Dunham, E. " Exploring Escape Fractions in ENZO Simulations of the Epoch of Reionization" (Submitted: The Astrophysical Journal).
- Wilken, S., Looney, L. W., **Satish, V**., Hussey, M. "Changing Opportunities to Enroll in Astronomy and Physics Courses in U.S. Public High Schools" (Submitted: APS Physical Review PER).

In Preparation

- Wilken, S., Looney, L. W., Talafian, H., **Satish, V**. " Motivations Behind Astronomy Instruction in Rural High Schools (In Prep: APS Physical Review PER).
- **Satish, V**., Barrow, K. S. S., Byrom, L. "Implementing a Redshift-Dependent IMF in ENZO Simulations" (In Prep: The Astrophysical Journal)

Work & Teaching Experience

Undergraduate Research Skills Bootcamp Working Group January 2026 – Present
Student Working Group Member

- Served as the sole undergraduate student representative in a working group developing a research skills bootcamp for undergraduate students.
- Contributed student perspectives to the development of active-learning activities and hands-on research curriculum.

Department of Astronomy, University of Illinois Urbana-Champaign August 2023 – May 2026
Telescope Observing Teaching Assistant

- Trained 30+ students on the operation and preservation of a national historic observatory telescope.
- Instructed new students on foundational astronomical concepts and observational techniques.
- Led 8 semesters of night observing and solar observing sessions for students in introductory astronomy courses, facilitating hands-on learning experiences.

Department of Astronomy, University of Illinois Urbana-Champaign September 2023 – May 2026
Course Grading Assistant

- Supported grading and instructional activities for ASTR 100: Introduction to Astronomy, ASTR 210: Introduction to Astrophysics, and ASTR 310: Computing in Astronomy, ASTR 330: Extraterrestrial Life, ASTR 406: Galaxies and Cosmology.

Campus Involvement

Astronomical Society, University of Illinois Urbana-Champaign April 2024 – May 2026
President

- Led planning and execution of Observatory Open Houses, new student events, and community star parties, fostering an inclusive and active astronomy community.
- Partnered with the Department of Astronomy to engage prospective students through admissions events, orientations, and outreach initiatives.
- Designed and ran outreach events for high schools, university students, and the general public.

Friends of the University of Illinois Observatory April 2024 – May 2026
Board Member

- Appointed as the Astronomical Society representative to advance preservation goals of the historic Campus Observatory.
- Initiated programs to strengthen alumni connections and increase community participation in observatory events and heritage efforts.

Illinois Space Society, University of Illinois Urbana-Champaign October 2022 – May 2024
Educational Outreach Board Member

- Supported STEM outreach initiatives connecting local schools with space and astronomy education opportunities
- Coordinated interactive events and workshops for middle and high school students to promote engagement in science and engineering.

Conference Participation

- **247th Meeting of the American Astronomical Society** | January 2026 | Phoenix, AZ
- **Astrofest 2026** | April 2026 | University of Illinois Urbana-Champaign
- **Undergraduate Research Symposium** | April 2025 | University of Illinois Urbana-Champaign
- **Astrofest 2025** | April 2025 | University of Illinois Urbana-Champaign

Technical Skills

Programming: Python, C++
Scientific Software: ENZO, FSPS, NumPy, yt, Healpy/HEALPix, Matplotlib